

36104 · CLASS 1

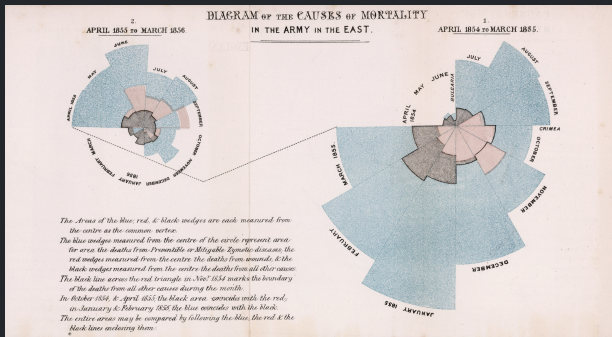
Seeing Data

Visualisation, intelligence, and interpretation

How do charts create meaning?

SILENT CHART READING · 5 MINUTES

Before anything is explained



Write down, without discussion:

- what you **see** — marks, colours, shapes;
- what you think it **means**;
- one thing you **cannot tell** from the image alone.

Keep this page. We return to it at the end of class.

Florence Nightingale, *Diagram of the causes of mortality in the army in the East*, 1858

(public domain).

Two kinds of sentence

- **“The chart shows ...”** — reserved for what is directly visible:
blue wedges dwarf red ones; the wedges shrink after March 1855.
- **“This suggests ...”** — interpretation:
disease killed more soldiers than wounds; sanitation reforms worked.

Most of what a chart “says” is supplied by the reader.

This class is about noticing when that happens — in you, and in a machine.

THE PREMISE

**A chart is a claim about
how the world is organised.**

Five decisions produce the claim

- **Selection** — which data, which window, which denominator.
- **Encoding** — which visual channels carry which variables.
- **Comparison** — what is placed next to what.
- **Framing** — title, caption, annotation, baseline.
- **Omission** — everything left out, silently.

Opening question for every artefact this semester:

what has the designer chosen to make visible — and what remains invisible?

LEARNING OUTCOMES

By the end of today

1. Explain how selection, encoding and framing turn data into meaning.
2. Separate observation from interpretation — in your own reading and in generated text.
3. Explain why identical summary statistics can conceal different structures.
4. Use an AI assistant to produce a chart, then verify every claim it makes.
5. Apply representation and governance questions to a chart critique.

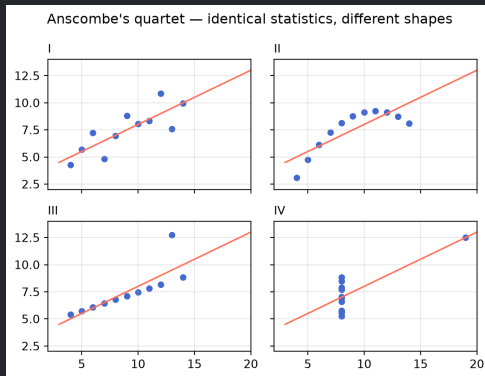
Four datasets, one set of statistics

	I	II	III	IV
mean of x	9.0	9.0	9.0	9.0
mean of y	7.50	7.50	7.50	7.50
variance of x	11.0	11.0	11.0	11.0
correlation	0.816	0.816	0.816	0.816
regression line	$y = 3.00 + 0.50x$			

Prediction: will the four datasets look essentially the same when plotted?
Write yes or no — and why — before the next slide.

ANSCOMBE'S QUARTET, 1973

Looking changes the conclusion



Course atlas reproduction of Anscombe (1973). A summary statistic is not a substitute for a picture.

The chain we will use all semester

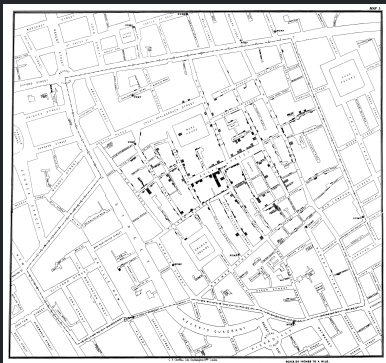
data → **encoding** → **observation** → **insight** → **narrative**

- **data** — records of the world, already shaped by collection;
- **encoding** — the mapping of values onto marks and channels;
- **observation** — what any careful reader can see (“the chart shows”);
- **insight** — an interpretation that survives verification (“this suggests, because”);
- **narrative** — insights arranged to support a decision.

Every failure we study this semester is a broken link in this chain.

WORKED EXAMPLE · LONDON, 1854

A picture can become an argument



Each black bar: one death,
stacked at an address.

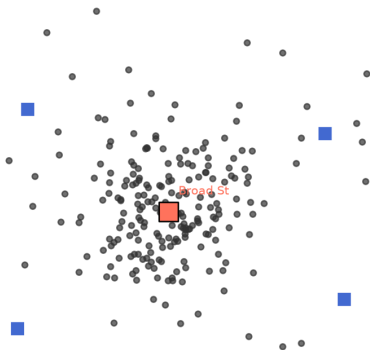
Read it with the chain:

- what is the **encoding**?
- what is the strongest **observation**?
- what **insight** is Snow selling?

John Snow's map of cholera deaths, Soho, 1854; published in *On the Mode of*

Pattern, inference, and the missing evidence

Snow's cholera map



- **Pattern:** deaths cluster around one pump. Visible.
- **Inference:** the water is the vehicle. Not visible — argued.
- **Causal evidence** came from elsewhere: the brewery workers who drank beer; the workhouse with its own well; deaths that stopped after the handle came off.

The map did not prove the theory.
It made the theory **impossible to ignore**.

Six ways a true chart misleads

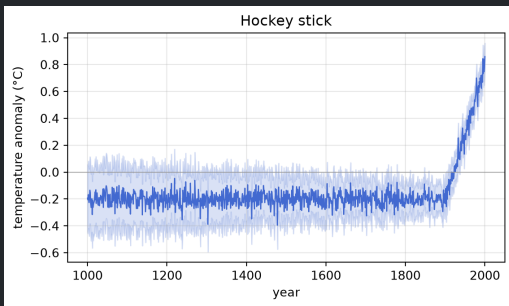
Every number checked, every label true — and the impression still wrong.

- truncated or missing baseline
- cherry-picked time window
- dual axes tuned to a story
- area standing in for value
- binning and smoothing choices
- omitted denominators

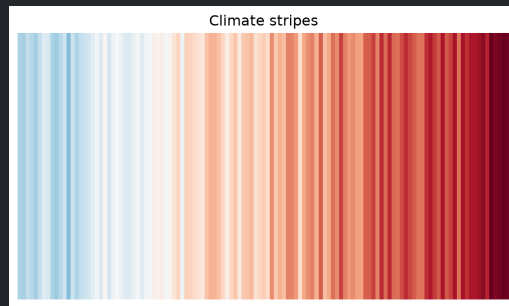
After Cairo, *How Charts Lie* (required reading, selected chapter).

The window is part of the claim

Crop either one to 1998–2012 and you can sell a pause. The axis window is an editorial decision.



Course atlas: the hockey stick — context makes the rise the story.



Course atlas: climate stripes — the same claim, at a glance.

VISUALISATION + AI

**Fluent interpretation is not
valid interpretation.**

What the assistant does well — and doesn't

- An AI system can describe a pattern **confidently** without checking whether the data supports the claim.
- Generated text is optimised to read well. Reading well and being right are different properties.
- The failure mode is not usually a wrong number — it is an **unsupported claim** delivered in a supported tone.

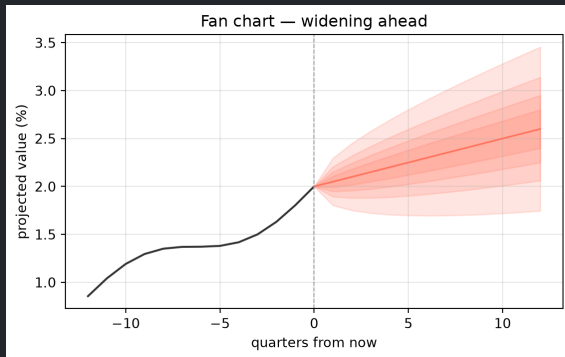
Your job is not to avoid the assistant. It is to **own the verification**.

The verification ladder

1. **Source** — what data was actually used? Can you point to it?
2. **Transformation** — what was filtered, aggregated, reshaped? Check counts and types.
3. **Visibility** — is each claimed pattern actually visible in the plotted data?
4. **Support** — classify each claim: **supported** / **plausible but unverified** / **unsupported** / **contradicted**.
5. **Disclosure** — record what the tool contributed and what remains unverified.

This ladder is the spine of the notebook today — and of Assessment 1, Part C.

Uncertainty drawn rather than suppressed



Course atlas: a fan chart admits what it does not know. Generated interpretations rarely volunteer this — you must ask.

5 MINUTES SOLO · 5 IN PAIRS · 5 AS A ROOM

Think · pair · share

Find a chart published **this week** — news, social, a report.

- What claim does it make?
- Which of the six mechanisms should worry you?
- What would you need to verify before repeating its claim?

Keep the chart. It may become your *Assessment 1* artefact.

BREAK · 10 MINUTES

**When we return:
who owns the data, then the lab.**

ACKNOWLEDGEMENT OF COUNTRY

Representation is also governance



This class was written and is taught on Aboriginal land.

More than 250 language, social and nation groups on one sheet — a claim that the land was never empty.

Its boundaries are deliberately soft: a general guide, contested in places, and explicitly not evidence for legal claims.

Aboriginal Australia Wall Map, D R Horton, © Aboriginal Studies Press, AIATSIS, 1996.

Four questions that outrank aesthetics

- Who is **represented** in this data — and who was consulted?
- Who **benefits** from this visualisation?
- Who could be **harmed or misled** by it?
- What limitation **must be disclosed** before it circulates?

Required reading: Maïam nayri Wingara & AIGI, *Indigenous Data Sovereignty Communique*. These four questions appear in every critique this semester, including Assessment 1.

Notebook lab · 45 minutes

1. **Anscombe** — summary statistics vs the picture.
2. **Generate and verify** — assistant-drawn chart of the transport dataset, then the checks: rows, types, missing values, totals.
3. **Claim audit** — a generated interpretation, every sentence classified: supported / plausible but unverified / unsupported / contradicted.

The contract: **predict** before you generate · give the assistant a **bounded task** · **verify** before you accept · **disclose** at the end.

Blocked on syntax? Ask for the starter notebook — that is what it is for.

Studio · Critique and Repair · 25 minutes

- **Read** the assigned artefact: main claim, audience, what is directly visible.
- **Test** the design: encodings, easy comparisons, what is aggregated or hidden.
- **Verify and repair**: what you would check, an alternative view, a repaired caption.
- **Responsibility**: who benefits — and who carries the cost if it is wrong?

Ten minutes solo, ten with a partner, five to revise.

Pair exchange · the Critique 1 frame

Swap artefacts: one **human-made chart**, one **AI-generated interpretation**.

- Same questions for both: claim, audience, task, comparison, omission.
- Then the AI-specific one: **which claims would survive the ladder?**

This is exactly Assessment 1, Parts A and C, at rehearsal scale:

Part A critique (supervised, in studio) · Part B redesign ×3 (next class) · Part C claim audit.

EXIT TICKET

Close the loop

Look back at your silent reading of the Nightingale diagram.

Which of your sentences were observations? Which were interpretations?

- One design decision I made today:
- One risk or limitation I noticed:
- One thing I need to verify before next class:

Diagnostic quiz on Canvas after class — open book, closed assistant, ungraded. It calibrates the course, not your mark. Next class: Choosing Visual Forms — what visual task are we performing?

Image credits and reading

- Nightingale (1858) and Snow (1855) images: public domain, via Wikimedia Commons.
- AIATSIS map: © Aboriginal Studies Press, AIATSIS and Auslig/Sinclair, Knight, Merz, 1996. Reproduced with attribution for teaching; view the official copy at aiatsis.gov.au.
- All other figures: course atlas reproductions (*How the World Is Organised*), generated from data.

Required reading: Cairo, *How Charts Lie* (ch.); D'Ignazio & Klein, *Data Feminism* (ch.); Maïam nayri Wingara & AIGI, *Indigenous Data Sovereignty Communique*.